# CS3 – BONE CAGE FOR SPINAL IMPLANTS

## PROJECT BRIEF:
To develop an interbody spinal cage to fuse the two vertebrae (spinal fusion) to treat back pain. The focus is to create a design that has an adaptive modulus, and bears an increasing load over time.

## DESIGN REQUIREMENTS:

### Patient Safety:

### Adaptive modulus
Try to align the temporal adaptive modulus with bone formation-Degradable materials

### Ability to bear an increasing load over time
Must be able to bear load until it is certain that new bone will have formed

### Drive bone formation
Include design features that do this-porosity, shape, material, surface charge.

### Radiolucent
Able to assess bone formation(allows x rays through so can see on scans)

## KEY INFORMATION

**Osteoconduction**
The ability of a material or scaffold to guide or facilitate the growth of new bone on its surface.-needs bone
Bone cells migrate to and proliferate along the surface, forming a scaffold for new bone formation.

**Osteointegration**
Integration of the implant with surrounding bone.

- **Ensures:** Implant and long-term stability

Bone Growth Process:
Osteoblasts migrate to the implant surface → form new bone → fuse with the implant material

**Osteoinduction**
The ability of a tissue or material to induce bone formation by stimulating undifferentiated cells to become osteoblasts-(bone growth where there isn't surrounding bone)

**Mechanics**
**Wolffs Law-**Bone adapts to the mechanical loads it experiences:

- With regular stress, bone remodels, becoming stronger and denser-stress encourages bone growth
- With reduced stress (e.g. from a stiff implant), bone weakens and loses density

**Stress shielding-** normal stress on bone is reduced because an overly stiff implant takes on a portion of the load-Leads to reduced bone density(osteopenia).
How to Reduce Stress Shielding:

- Use more flexible or graded-stiffness materials
- Incorporate porous or lattice designs
- Match implant stiffness to surrounding bone (biomimetic design → modulus compatibility)

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## PATIENT SAFETY

Objective: Ensure safe load-bearing until bone is sufficiently formed.

**Example q: Discuss how material and structural design choices in spinal cages ensure patient safety and effective spinal** fusion Answer structure: Material selection, Mechanical performance, Degradation risks, Patient-specific design (slot-in inserts, adaptive modulus)

### BIOMIMETIC CAGE LAYOUT-LOAD DISTRIBUTION+MECHNAICAL COMPATIBILITY

Material strategy: Use two complementary materials to overcome the limitations of using only one.

**Outer Layer- Permanent, Hard support (Ti / $Si_3N_4$ / Coated PEEK)**

- **Chosen material: 3D printed Ti**-high porosity-post processing to get submicron features on outside
- **Placement:** On the apophseal ring, which bears the most load and is made of very **strong cortical bone**→ Therefore,hard material like Ti is used to **match the modulus** and provide **mechanical compatibility.**

**Inner layer- Soft, resorbable support**

- **Material: PART Mg alloy**, aligned with **cancellous bone** in vertebral center
- **Placement:** Slot-in design allows **personalisation** at point of use
    - Surgeon selects fast/medium/slow resorbing insert during surgery-**matching likely biological response to the patient**

➤ **Adaptive Modulus:** Mg gradually degrades, transferring stress to new bone over time

- ➤ **Key Safety Need:** Degradation must be synchronised with bone growth→**Coating (e.g., Chitosan-** Slows Mg degradation to prevent premature mechanical loss (-Mg likely to degrade too quickly for new bone growth)

**Bone Graft Chamber-Central slot**

Central slot allows addition of bone graft material like rhBMP-2 at point of surgery

**Roles:**

1. **Stimulate Bone Growth** – e.g., BMPs trigger natural bone-forming pathways
2. **Provide Structural Support** – Scaffold for osteoblast attachment
3. **Promote Fusion** – Facilitates formation of solid bone bridge between vertebrae

**Additional Design Features**

- **Porosity & Lattice Structures:**
    - Reduce **stress shielding**
    - Improve **bone integration** and nutrient/waste flow
- **Antimicrobial Coatings:**
    - **Silver:** Prevents biofilm, shown to be **safe for bone regrowth**
    - **Copper:** May boost angiogenesis via hypoxia-mimicking signals

## MATERIAL COMPARISON

Inner Layer: Mg Alloy vs PLA

- **Mg Alloy:**
    - Newer in some evaluation shows it upregulates/contributes to bone formation
    - **Less radiodense** than PLA → doesn't show up as much on X-rays

      ➤ Advantage: as Mg degrades, **bone growth becomes more visible**, helping assess healing
    - **Risks:**
        - Can degrade too fast if uncoated
        - Potential cytotoxicity if too much is used
        - Requires further validation — best used in **part**, not full structure
- **PLA (Poly-lactic acid):**
    - Long clinical history
    - **More radiodense** → shows up well on scans, but can **obscure bone growth**
    - Acid buildup from degradation accelerates breakdown → feedback loop
    - Osteoblasts **don't prefer PLA** compared to Mg(cells don't like to grow on it)

- Resorption is **variable and pH-sensitive**

## Outer Layer: Ti vs PEEK vs CFR-PEEK

- **Titanium (Ti):**
  - High modulus → strong support
  - Easy to post-process (acid etch) → **nanoscale surface** features- nanoscale features easier to do on titanium can do post processing like acid etching,alkaline etching etc-create very fine pores and grains-which osteoblasts adhere to
  - Best for screw fixation + resisting mechanical failure
- **PEEK:**
  - Radiolucent and lightweight
  - **Drawbacks:**
    - Hydrophobic → poor cell adhesion
    - Hard to form pores or concave shapes
    - Cells don't like to grow on PEEK like they do on Ti.
    - Often used for cost savings -harder to create right size and shape of pores with peek than it is with Ti-which is why market has generally moved away from peek-60% Ti 40% Peek-a lot of the people who are using it are because its slightly less costs-so in terms of manufacturing=less costs for hospital
- **Carbon Fibre-Reinforced PEEK (CFR-PEEK):**
  - Can't be 3D printed easily
  - Printing risk: fibre fragments stick out → **risk of migration in body**
  - Only safe if **fully embedded**

## Mechanical Concerns with Polymers

- **Polymers like PEEK:**
  - Easier to deform over time, especially before fusion is complete
  - Poor screw holding strength
  - Degradation may reduce mechanical integrity before bone healing finishes
- **Ti Advantage:**

  Stiff, durable, consistent under force- typically screws are put to stabilise cage-with polymer difficult as they are softer and weaker -undergoing that force over time-TI much better to resist that force

  - Handles long-term fixation and dynamic loading better

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## BONE FORMATION

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Example question- **Explain how design features of spinal cages promote bone growth."**
**Answer Structure:** Porosity + surface shape(concave) + coating + graft + charge

- **Porous Structure:** Allows nutrients, blood vessels, and bone cells to move through the cage.(Especially needed for when material degrade-it needs to move through and cant build up)

### Concave Pores: Concave Pores (Biomimetic Cues):

- Mimic osteoclast resorption pits.
- Osteoblasts prefer curved, concave areas — *despite Wolff's Law suggesting cell migration toward load, osteoblasts are observed to attach and grow in unloaded concave pits.*
- Bone remodels frequently: osteoclasts dissolve bone → create concavities → osteoblasts fill in.
- **Concave pores** are essentially **biomimetic repair sites**.

**Submicron/Nanoscale Features:** Increase surface area→ Enhance **protein adsorption** and **attracts cells**→ Promote **osteoinduction**

**Evidence:**

- **Calcium phosphate study:**
  No bone growthon flat sites. However, when only the **surface features** were modified—adding **submicron pores** and creating **smart concavities** (small, curved holes)—bone formation occurred in **muscle pouches** (non-bony sites).
- **Picard et al. (2023):**
  Used **titanium tubes** with concave surfaces and found that **bone formed without any surrounding bone** or chemical additives=shape driven osteoinduction
- **Surface shape alone** — especially **concave geometry** — was enough to trigger bone formation

**Surface Coatings:**

- **HA on Ti/PEEK** → promotes bone on-growth
- **Nanoscale Ti** on PEEK enhances bioactivity.(easier to do with Ti)-get the benefit of cell integration onto titanium without the disadvantages of it
- Permanent surface charge: Attracts and binds endogenous growth factors to the surface

**Biological Additions:**

- **rhBMP-2** in central chamber encourages new bone formation (osteoinductive)

**Piezoelectric Materials (e.g., PVDF):** Create small currents during movement — shown to upregulate bone formation-this material can be used as a coat-when patients walking that deformation creates small current-attracts osteoblasts

**Use of Antimicrobial Coatings: Silver & Copper**

Silver:

- Effective at preventing biofilm formation (antibacterial)
- Concern: COULD IT NEGATIVELY IMPACT OSTEOBLAST ACTIVITY?
- Evidence suggests silver is safe — does not inhibit bone regrowth when used at appropriate concentrations
- Already used in several approved devices (e.g., dental implants, trauma plates)

Copper:

- Less commonly used but being explored
- May stimulate angiogenesis by mimicking hypoxic signals
- Unclear how copper directly affects osteoblast activity — needs more research
- Speculative benefit in promoting vascularised bone healing

## RADIOLUCENCY

- All polymer-based materials might offer advantages over metals or other radiopaque materials
- Better still to use adaptive radiolucent – a design that is initially radiopaque, but the resorbing elements would allow increase visualisation of new bone formation over time

## Manufacture:

3D-printed titanium spinal cages are typically manufactured using **Selective Laser Melting (SLM)** or **Electron Beam Melting (EBM)**, which build the implant layer by layer from titanium powder. SLM uses a high-powered fiber laser (~1060 nm) to melt the powder, while EBM uses a high-voltage (60–150 kV) electron beam in a vacuum for improved material properties. After printing, **post-processing** includes powder removal, heat treatment to relieve stress, and surface treatments to enhance bone integration. Implants then undergo **quality control** through mechanical and dimensional testing. Finally, they are **sterilized** using gamma radiation, autoclaving, or ethylene oxide gas and packaged for surgical use.

**SLS:** Uses a laser to **fuse powder without fully melting it** (mostly polymers, some metals). **Limitations:** Not ideal for high-strength implants like spinal cages. **Preferred Alternatives: SLM/EBM** melt the material completely → better for **dense, load-bearing parts.**